



GEN AI KICKSTARTER · PROJECT 01

SOLUTION

Personal Research Agent

An always-ready research agent that answers across your PDFs, links, and videos — with a source behind every line.

Day 5 · Perplexity · NotebookLM · Claude

Reference build — yours can differ.



Personal Research Agent

How it's executed

THE EXECUTION — HOW, WHERE & WHEN

You run two grounded engines and one synthesiser. First, NotebookLM becomes your private library: you add every source and ask questions answered only from that material, each line linked to its passage. Second, Perplexity covers anything newer, returning live answers with numbered citations. Finally, Claude merges the two into a single one-page brief, flags where sources disagree, and keeps every citation. About 30 minutes to set up; seconds to use afterwards.

WHAT TOOLS — AND THEIR ROLE

NotebookLM	Grounds answers in YOUR sources; auto-transcribes YouTube; cites exact passages
Perplexity	Live, web-grounded answers with numbered sources for anything recent
Claude	Synthesises both into one cited brief; becomes the reusable agent

TOOL OVERVIEW

What you'll find *where*

NotebookLM

Sources panel (left)	"+ Add" takes PDFs, links, YouTube URLs, Google Docs/Slides, or pasted text (up to 50 sources)
Chat (centre)	Ask anything; answers show citation chips you click to jump to the source
Studio (right)	Audio Overview (a podcast of your sources) + saved notes, FAQ and study-guide generators

Perplexity

Search bar (top)	Ask; choose a Focus (Web, Academic...) and attach files
Sources strip	Numbered [1][2] citations under every answer — click to open
Spaces (left)	Save one topic's files and threads together so context persists

MAKE IT REUSABLE

Connect it & automate it

The build is worth more when it plugs into your real tools and runs itself. Here's exactly where to click.

CONNECT A CONNECTOR

Where: Customize → Connectors (or "+" in the message box → Connectors → Manage)

- 1 Pick the app from the directory — Google Drive, Gmail, Notion and more.
- 2 Click Connect and approve in the app's own sign-in window (OAuth).
- 3 Grant the least access it needs; you can turn it on per chat.
- 4 In a chat, toggle it on and ask Claude to use it.

For this project: Connect Google Drive so NotebookLM and Claude read your saved PDFs and notes directly.

Claude never sees your password — and it never sends or changes anything until you say yes.

CREATE A SKILL

Where: Customize → Skills · in Cowork: "+" → Record a skill

- 1 Record: "+" → Record a skill → do the task once → Save — Claude writes the SKILL.md for you.
- 2 Or write it: a SKILL.md with a name, a one-line description (that's the trigger), and 5–8 rules.
- 3 Zip the folder (folder as the root) → Customize → Skills → Upload → enable.
- 4 Run it with /your-skill; edit and re-upload to improve it.

For this project: Save your synthesis prompt as /research-brief so any new question runs the full pipeline.

A Skill is a saved recipe Claude reloads every time — so you never re-explain.

PROMPT LIBRARY · STAR

Write it in *STAR*

STAR = Situation · Task · Action · Result. Same four parts every time — start at your level and add detail as you climb.

BASIC

- S Situation.** I've collected a few sources on [topic].
- T Task.** Summarise the main points.
- A Action.** Read all of them and group similar ideas.
- R Result.** 5 plain-English bullets.

INTERMEDIATE

- S Situation.** My notebook holds 8 sources on [topic].
- T Task.** Pull the findings I can trust.
- A Action.** Answer only from my sources; attach a citation to each point.
- R Result.** 5 findings, each with its source and passage.

ADVANCED

- S Situation.** I have two cited answers on [topic] — one from my notebook, one from live web.
- T Task.** Merge them into one brief I can act on.
- A Action.** Combine both, flag where they disagree, drop anything uncited, keep [n] markers.
- R Result.** A one-page brief: 3–5 findings, the disagreements, and a short “what I’d do”.

PRO

- S Situation.** I run a standing research agent for [topic] I'll query often.
- T Task.** Answer any new question reliably and tell me what I'm missing.
- A Action.** Answer from my notebook with citations; say if it needs fresh data and what to search; rate confidence; list gaps.
- R Result.** A cited answer + confidence + the 3 sources to read next. (Save as /research-brief.)

WHY IT MATTERS

Where this pays off *in real life*

This isn't a toy exercise. The same build, pointed at real work, does this:

- ✓ Exam & interview prep from your own notes
- ✓ Market and competitor research before a decision
- ✓ Due diligence on a company or product
- ✓ Literature reviews and reports
- ✓ Staying current in a fast-moving field

WHAT GREAT LOOKS LIKE

- ✓ Every claim in the brief is cited — no naked assertions
- ✓ Disagreements between sources are surfaced, not hidden
- ✓ You can answer a new question on the topic in under a minute